Optimizing Surgical Patient Flow through Operating Room Scheduling: A Use Case Study

Walid Abdelaidoum [1,2], Mohamed A. Madani [2], Larbi Boubchir [1], Boubaker Daachi [1], Louis Dagneaux [3] and Stanislas de Coüasnon[2]

[1] *LIASD, University of Paris 8, Paris, France*

[2] *SWAPPY, Paris, France*

[3] *Hôpital Lapeyronie, Orthopedic Service, Montpellier, France*

walid.abdelaidoum@etud.univ-paris8.fr, abdou@swappy.fr, larbi.boubchir@univ-paris8.fr, professeur.dagneaux@gmail.com, sdecouasnon@swappy.fr

***Abstract*—Operating room (OR) scheduling often prioritizes operational efficiency over patient preferences. In our study, we present a patient-centered scheduling framework developed for the orthopedic department at Hôpital Lapeyronie, the University Hospital of Montpellier, France. Formulated as a mixed-integer linear program (MILP), the model handles strict resource constraints with patient availabilities. This framework generates a set of optimized time slots, giving the patients the ability to select their preferred surgery date. We used an iterative heuristic that proposes a patient at a time, where the physician sets with the patient the best schedule date between three options, allowing the excluded ones to be considered for the other patients. The results using use case data show that the MILP model increases the theoretical capacity of the operating rooms utilization by 23%. Benchmarks were also generated to simulate daily uses and showed that, on average, more than 2.5 slots are proposed to each patient. This tool shows that efficiency can be achieved with humane consideration by integrating operational objectives with patient satisfaction.**

***Index Terms*—Operating Room Scheduling, Surgical Patient Flow, Mixed-integer Linear Programming, Optimization, eHealth.**

I. Introduction

Operating room scheduling is a research domain that continues to attract significant interest, with ongoing novelty in methodological approaches and practical applications. It is classified as a multi-objective optimization problem. The objectives are first to minimize costs: in this study [4] surgical cate accounts for nearly one third of US health care spending. And second, to maximize operational utilization, considering the hospital's operational constraints and operation types, while also accounting for the comfort of medical staff and patients. Most studies focus on operational aspects and propose deterministic and stochastic optimization models, which generate a predetermined schedule without taking patient choice into account. This is understandable for certain types of operations, such as cardiovascular or emergency cases, where the intervention should be performed as soon as possible.

However, this study is set in the orthopedic department at the Centre Hospitalier Universitaire (CHU) Montpellier (France), which provides a suitable environment for developing a patient-centered scheduling approach. Indeed, unlike emergency or time-critical surgical cases, elective orthopedic patients generally have more flexibility in their scheduling preferences, even when urgent cases are involved. This flexibility enables the integration of patient preferences into scheduling decisions. This work aims to balance operational efficiency with patient care, enabling patients to actively participate in the scheduling process.

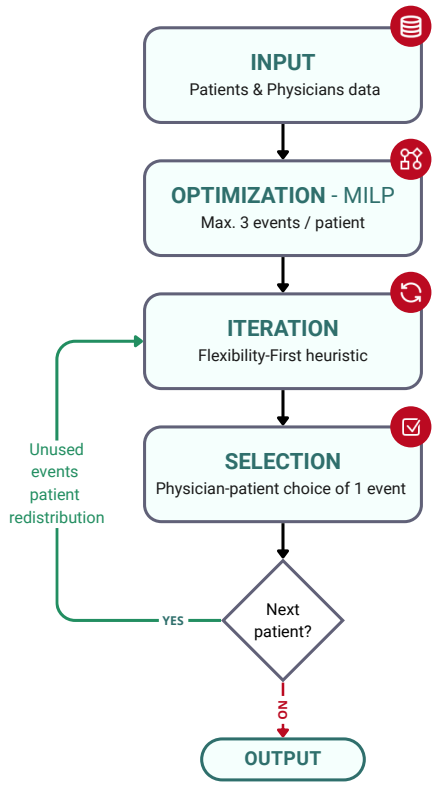

FIG. 1: Iterative scheduling process: MILP generates feasible events, the flexibility-first heuristic selects and confirms patient selections, and unused events are redistributed.

As shown in Figure 1, this approach is based on an iterative optimization methodology. First, patient and physician data are uploaded into the system. Based on this dataset, the MILP model generates up to three slots that match the availability of both patients and physicians. Next, the flexibility-first

heuristic identifies the patient with the most flexible schedule, as determined by the quality scores σ_p (detailed in Subsection III-D). The selected patient reviews the available options and chooses the one that works best for them. Then, the unused options are redistributed to the remaining patients, and the system recalculates which patient to process next until there are no patients left to assign. By actively including the patients in the scheduling system, this iterative method combines both efficiency and care.

II. Related Work

Operating room scheduling has been extensively studied across multiple domains [2], with various techniques being applied to address its inherent complexity. Among heuristic approaches, Kamran et al. [8] applied a column-generation-heuristic-based Benders' decomposition on adaptive allocation scheduling of patients in operating rooms, while Wang et al. [15] developed a Wasserstein distributionally robust optimization model with a rolling horizon rescheduling scheme for surgery scheduling involving both elective and emergency patients. In the field of artificial intelligence, recent studies have leveraged reinforcement learning and machine learning techniques to tackle scheduling challenges. Lex et al. [9] applied a two-stage predict-then-optimize approach combining machine learning prediction with scheduling optimization for elective orthopedic surgery. Similarly, Zhao et al. [18] introduced a hybrid framework integrating multi-level optimization with reinforcement learning and column generation for operating room scheduling. El-Balka et al. [6] proposed a dynamic operation room scheduling (DORS) strategy combining explainable AI for feature selection with a fuzzy interface engine for intraday surgical scheduling. Furthermore, Eshghali et al. [7] applied a random forest machine learning model combined with genetic algorithm and particle swarm optimization for integrated scheduling and rescheduling of elective and emergency patients. Several studies have also adopted stochastic approaches to handle inherent uncertainties in surgical scheduling [3]. M'Hallah and Visintin [11] applied a stochastic integer programming model with sample average approximation for scheduling elective surgeries in a cyclic Master Surgical Schedule, whereas Zhang et al. [17] developed a two-phase optimization model combining Markov decision process and stochastic programming for advance surgery scheduling. Exact solution methods have also been widely explored in this domain. Çelik et al. [19] applied a two-stage stochastic mixed-integer programming model with a progressive hedging algorithm for surgery scheduling under the parallel processing principle. Schneider et al. [14] employed a MILP approach for scheduling surgery groups while considering multiple downstream resources. Miao and Wang [12] proposed a MIP model with a preventive-reactive scheduling procedure for elective and emergency surgery scheduling with waiting time limits. Akbarzadeh and Maenhout [1] developed a dedicated branch-price-and-cut algorithm for simultaneous patient planning and surgeon scheduling in operating rooms. Additionally, Makboul et al. [10] applied a multiobjective ϵ-constraint based robust optimization approach to the master surgical schedule problem under multiple uncertainties, including surgery duration and downstream resource constraints. Other relevant studies have further contributed to this growing body of literature [16, 5].

This study is directly relevant, Spratt and Kozan [13] proposed constructive heuristics with real-time reactive strategies for the multiple operating room surgical case sequencing problem. Additionally, this heuristic approach achieves real-time performance. In this paper [5], the authors have applied a disruption management model based on distributionally robust optimization and machine learning consensus on integrated scheduling for inpatient surgery and ambulatory surgery; they found that their approach effectively coordinates shared surgeons across different operating rooms while handling uncertainty in surgery duration and disruptions.

In this study, we focus on two main objectives. The first is to propose an optimized schedule that minimizes operational costs and maximizes block utilization while respecting all hard constraints. The second is to prioritize patient choice and flexibility by proposing two to three distinct feasible slots per patient, taking into consideration all patient availabilities and guaranteeing at least one feasible slot for every patient.

III. MILP Model with Patient Preference

A. Operational Context

The scheduling operates within a clinical workflow that differs from traditional batch scheduling approaches. In practice, patients are not accumulated and scheduled in large batches at fixed weekly intervals. Instead, the system processes patients incrementally as they complete their consultation appointments, where the operating physician determines the surgical need and assigns the patient to their surgical schedule.

In this incremental process:

- Patients enter the scheduling system immediately after consultation.
- The optimization model runs for each new patient (or small groups of patients from a single consultation day).
- The system generates up to three optimized feasible time slots tailored to each patient's constraints.
- Patients review these options and confirm their preferred surgery date.
- Once confirmed, the assignment becomes fixed in the schedule for subsequent optimization runs.

Consequently, each physician maintains a wait list of patients pending surgical scheduling, which typically contains between 5 and 15 patients at any given time. This wait-list represents patients who have been consulted and are awaiting either (a) the generation of feasible time slots, (b) patient confirmation of their selected slot, or (c) resolution of specific medical or administrative prerequisites.

Wait-lists remain small (5-15 patients, maximum 20) because patients flow continuously through the system rather than accumulating into backlogs.

Symbol	Description
$\mathcal{D}$	Set of planning days, index d
$\mathcal{B}$	Set of operating rooms (blocs), index b
$\mathcal{H}$	Set of physicians, index h
$\mathcal{P}$	Set of patients, index p
$\mathcal{I} = \{0, \ldots, I_{\max} - 1\}$	Set of proposed time slot option indices for each patient, index i
$\mathcal{K}$	Set of operation types, index k
$\mathcal{D}^{\text{part}}$	Family of day partitions (for spacing constraint), each partition $P \in \mathcal{D}^{\text{part}}$ is a subset of $\mathcal{D}$
τ_p	Operating time of patient p
τ_p^{post}	Post-operative time of patient p
π_p	Priority of patient p
$k(p) \in \mathcal{K}$	Operation type of patient p
d_p^{arr}	Earliest feasible day index for patient p
d_p^{dep}	Latest feasible day index for patient p
T	Planning horizon length (in days); $\mathcal{D} = \{0, \ldots, T-1\}$
stdPost	Standard post-operative buffer time (scheduling flexibility allowance added to block capacity)
$B_{b,d} \in \{0,1\}$	Indicator: block b is open on day d (when $B_{b,d} = 0$, capacity is zero)
$C_{b,d}$	Available OR-time on (b, d)
$\overline{E}_{b,d}$	Upper bound on capacity violation slack
$w_{p,h}$	Compatibility weight between patient p and physician h
$a_{p,h,b,d}$	Preference weight for assigning (p, h) to (b, d)
$n_{b,d}^{\text{pre}}$	Number of pre-defined operation types in block b on day d (aggregated across all physicians)
M	Big-M constant, e.g. $M \geq \lvert\mathcal{B}\rvert \cdot \lvert\mathcal{H}\rvert \cdot \lvert\mathcal{P}\rvert \cdot \lvert\mathcal{I}\rvert$
$\alpha_1 = 10^3$	Weight of patient-assignment errors
$\alpha_2 = 10^1$	The weight of the block capacity violation (is only used to schedule urgency, replacing elective patients).
$\alpha_3 = 10^1$	Weight of early/late scheduling w.r.t. stay
$\alpha_4 = 10^{-1}$	Weight of specialty fragmentation
$\alpha_5 = 10^{-3}$	Weight of day usage

TABLE I: Data description (sets and parameters).

Variable	Domain	Description
$X_{d,b,h,p,i}$	$\{0,1\}$	1 if patient p is scheduled on day d in block b with physician h and index i
z_d	$\{0,1\}$	1 if day d is used (at least one assignment), 0 otherwise
$e_{p,i}$	$\{0,1\}$	1 if patient p is not assigned at index i (slack for one-time assignment)
$E_{b,d}$	$\mathbb{Z}_{\geq 0}$	Capacity violation (extra time) allowed for block b on day d
$EA_{p,i}$	$\{0,1\}$	Early-arrival violation indicator for patient p at index i
$LD_{p,i}$	$\{0,1\}$	Late-departure violation indicator for patient p at index i
$N_{b,h}$	$\mathbb{Z}_{\geq 0}$	Upper bound on the number of distinct operation types for physician h in block b on any single day
$u_{d,b,h,k}$	$\{0,1\}$	1 if physician h performs operation type k in block b on day d

TABLE II: Decision variables.

B. Constraints description

We formulate the scheduling decision as an optimization problem satisfying multiple constraints and optimizing a hierarchical multi-objective function (data notations are in Table I):

Clinical and Temporal Constraints: The solution must respect essential patient-specific requirements, including the operating time (τ_p), post-operative time (τ_p^{post}), and the patient's earliest (d_p^{arr}) and latest (d_p^{dep}) feasible scheduling days. Furthermore, the schedule must adhere to the resource availability (OR opening $B_{b,d}$, capacity $C_{b,d}$) and the compatibility between patients and physicians ($w_{p,h}$).

Hierarchical Objectives: The objective minimizes a weighted sum of violations, structured hierarchically by penalty weights (α_1 to α_5):

- **Priority 1:** Minimize unassigned patients (α_1).
- **Priority 2:** Minimize capacity and time constraint violations, including early or late scheduling relative to the patient's stay limits (α_3). Capacity violations (α_2) are permitted only for urgent cases replacing elective patients.
- **Priority 3:** Minimize specialty fragmentation (α_4) and efficiently manage day usage (α_5), which promotes operational stability and maximizes resource utilization.

C. Mathematical model

The operational scheduling problem is formulated as a MILP, with the objective to assign patients across operating rooms and days. The model's sets and parameters are detailed in Table I, and its decision variables are defined in Table II.

1) Objective: The objective function maximizes Z using hierarchically ordered weights (α_k). Patient assignment receives highest priority: the model minimizes assignment errors ($\alpha_1 = 10^3$, $e_{p,i}$) while maximizing rewards for high-priority patients (π_p), preferred physician matches ($w_{p,h}$), and favorable block-day assignments ($a_{p,h,b,d}$). Earlier intra-day scheduling (smaller index i) receives proportionally higher reward.

$$\max\ Z = + \sum_{d,b,h,p,i} X_{d,b,h,p,i} \Big[\pi_p(T-d) + w_{p,h} + a_{p,h,b,d}\Big](i+1)^{-1} \quad (1)$$

$$- \alpha_1 \sum_{p,i} \pi_p\,(i+1)^{-5}\, e_{p,i} \quad (2)$$

$$- \alpha_2 \sum_{b,d} E_{b,d} \quad (3)$$

$$- \alpha_3 \sum_{p,i} (EA_{p,i} + LD_{p,i}) \quad (4)$$

$$- \alpha_4 \sum_{b,h} N_{b,h} \quad (5)$$

$$- \alpha_5 \sum_{d} d\, z_d \quad (6)$$

2) Constraints: **Patient one-time assignment (with slack).**

$$\sum_{d\in\mathcal{D}} \sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} X_{d,b,h,p,i} \ + \ e_{p,i} = 1, \qquad \forall p \in \mathcal{P},\ \forall i \in \mathcal{I} \quad (7)$$

(If $e_{p,i} = 1$ then p is not assigned at index i; otherwise p must be assigned exactly once at that index.)

Daily block capacity constraints (with slack).

$$\sum_{h\in\mathcal{H}} \sum_{p\in\mathcal{P}} \sum_{i\in\mathcal{I}} X_{d,b,h,p,i}\,(\tau_p + \tau_p^{\text{post}}) \le C_{b,d} + \text{stdPost} + E_{b,d} \quad (8)$$

$$0 \le E_{b,d} \le \overline{E}_{b,d}, \qquad \forall b \in \mathcal{B},\ \forall d \in \mathcal{D} : B_{b,d} = 1 \quad (9)$$

Patient availability vs. arrival/departure (early/late violations).

Let d_p^{arr} and d_p^{dep} be the first/last admissible day indices for patient p.

$$\sum_{\substack{d\in\mathcal{D}:\\ d<d_p^{\text{arr}}}} \sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} X_{d,b,h,p,i} = EA_{p,i}, \quad \forall p \in \mathcal{P},\ \forall i \in \mathcal{I} \quad (10)$$

($EA_{p,i}$ may be fixed to 0 if early reprogramming is not allowed for p).

$$\sum_{\substack{d\in\mathcal{D}:\\ d>d_p^{\text{dep}}}} \sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} X_{d,b,h,p,i} = LD_{p,i}, \quad \forall p \in \mathcal{P},\ \forall i \in \mathcal{I} \quad (11)$$

($LD_{p,i}$ is fixed to 0 if late reprogramming is not allowed for p).

Day-activation constraints.

$$\sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} \sum_{p\in\mathcal{P}} \sum_{i\in\mathcal{I}} X_{d,b,h,p,i} \le M \cdot z_d, \qquad \forall d \in \mathcal{D} \quad (12)$$

If any $X_{d,b,h,p,i} = 1$ for a given d, then z_d must be 1 (and the day is penalized in the objective by (6)).

Specialty indicators and fragmentation per doctor and room.

$$u_{d,b,h,k} \ge X_{d,b,h,p,i}, \qquad \forall p \in \mathcal{P} : k(p) = k, \quad \forall i \in \mathcal{I} \quad (13)$$

$$u_{d,b,h,k} \le \sum_{p\in\mathcal{P}:k(p)=k} \sum_{i\in\mathcal{I}} X_{d,b,h,p,i} \quad (14)$$

so $u_{d,b,h,k}$ becomes 1 if (b,h) actually executes at least one case of type k on day d.

The aggregated number of specialties per physician and block is constrained by

$$\sum_{k\in\mathcal{K}} u_{d,b,h,k} - 1 + n_{b,d}^{\text{pre}} \le N_{b,h}, \quad \forall d \in \mathcal{D}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H} \quad (15)$$

Minimum spacing between slots per patient

$$\sum_{d\in P} \sum_{b\in\mathcal{B}} \sum_{i\in\mathcal{I}} X_{d,b,h,p,i} \le 1, \quad \forall P \in \mathcal{D}^{\text{part}}, \forall h \in \mathcal{H}, \forall p \in \mathcal{P} \quad (16)$$

With partition interval equal to 1 in the code, this effectively prevents multiple appointments of the same (p,h) within the same partition interval.

D. Flexibility-First Heuristic

Patients select their surgery time slots sequentially rather than simultaneously. Once a patient confirms a time slot, it becomes fixed, and subsequent optimizations will not take it into consideration for further recalculations. This reduces the possible options for patients remaining in the pool. Therefore, the order in which patients are presented for selection is critical to maximizing the feasibility of the solution and preserving the best options for all patients.

This heuristic prioritizes patients with the most flexible schedules (high availability scores) based on the MILP results for early selection. Such flexible patients can choose from multiple good-quality slots. For each unscheduled patient pp p, an availability score σ_p quantifies scheduling flexibility:

$$\sigma_p = \sum_{s\in\mathcal{S}_p} \text{score}(s,p) \quad (17)$$

where $\mathcal{S}_p$ is the set of feasible slots and $\text{score}(s,p)$ is:

$$\text{score}(s,p) = \begin{cases} +100 & \text{if } s \in [d_p^{\text{arr}}, d_p^{\text{dep}}] \\ -|s - d_p^{\text{arr}}| & \text{if } s < d_p^{\text{arr}} \\ -|s - d_p^{\text{dep}}| & \text{if } s > d_p^{\text{dep}} \end{cases} \quad (18)$$

Patients with higher availability scores σ_p are prioritized in the selection sequence. After each patient confirms a selection, slot availability is updated and scores are recomputed for remaining patients. Algorithm 1

Algorithm 1 Iterative Patient Selection and Confirmation

Input: Proposed slots $\mathcal{S}_p$ for each patient $p \in \mathcal{P}$, current assignments $A \leftarrow \emptyset$
Output: Confirmed assignments A^*
Initialize $A^* \leftarrow \emptyset$, $\mathcal{P}^{\text{remain}} \leftarrow \mathcal{P}$
while $\mathcal{P}^{\text{remain}} \neq \emptyset$ **do**
  for each patient $p \in \mathcal{P}^{\text{remain}}$ **do**
    Compute score: $\sigma_p \leftarrow \sum_{s \in \mathcal{S}_p} \text{score}(s, p)$
  end for
  $p^* \leftarrow \arg\max_p \sigma_p$ (select most flexible patient)
  Sort $\mathcal{S}_{p^*}$ by $\text{score}(s, p^*)$ in descending order
  Display sorted slots to user (physician or patient)
  Receive user confirmation: $s^* \in \mathcal{S}_{p^*}$
  $A^* \leftarrow A^* \cup \{(p^*, s^*)\}$
  $\mathcal{P}^{\text{remain}} \leftarrow \mathcal{P}^{\text{remain}} \setminus \{p^*\}$
  Re-optimize: Update $\mathcal{S}_p$ for all $p \in \mathcal{P}^{\text{remain}}$ given fixed assignments A^*
end while
return A^*

IV. Numerical Results

A. Hospital Data Structure

The computational experiments use data from the Department of Orthopedic Surgery at CHU Montpellier for elective surgical procedures over a three-week period. This framework is used to test the model's ability to optimize and condense the pre-existing surgical schedule.

The dataset includes 134 patients requiring surgery, categorized into four main procedure types: arthroplasty, foot and ankle surgery, soft tissue repair, and arthroscopy.

Operative time ranges from 15 to 240 minutes, with mean duration 71.86 minutes and median 80 minutes. A fixed 50-minute post-operative turnover time is assumed for all cases.

A total of 11 physicians are involved. Their schedules are fixed, requiring availability in at least one of the two weeks modeled. Two head physicians are included; one is capable of managing "swinging rooms," operating concurrently in up to 3 Operating Rooms (ORs).

Three OR blocks are available all working weekdays, providing 10.5 hours of capacity daily. One limited OR is open only on Wednesdays and Fridays, providing 6.5 hours of capacity daily.

B. Real-World Performance

The results, summarized in Table III and Figure 2, demonstrate the model's capacity to improve resource deployment while maintaining operational flexibility. To isolate maximum resource utilization, tests limited each patient to a single scheduled event. This step directly assessed the theoretical capacity within existing real-world resource constraints.
The optimization model demonstrates strategic redistribution across the planning horizon. In Weeks 1–2, utilization increased from 72–73% to 91–95% by condensing workload into available block time. In the third week, utilization decreased from 79% to 3% because most patients had already been assigned to the first two weeks. In these tests, we did not include fictitious patients; the objective was to redistribute existing patients to evaluate whether utilization gains were achievable in previously completed schedules.

TABLE III: Comparison of Surgical Block Time Utilization (%): Real Case vs. Optimized Schedule over 3 Weeks

Week	Real Case (%)	Optimized (%)	Change (p.p.)
1	72%	95%	+23
2	73%	91%	+18
3	79%	39%	-40

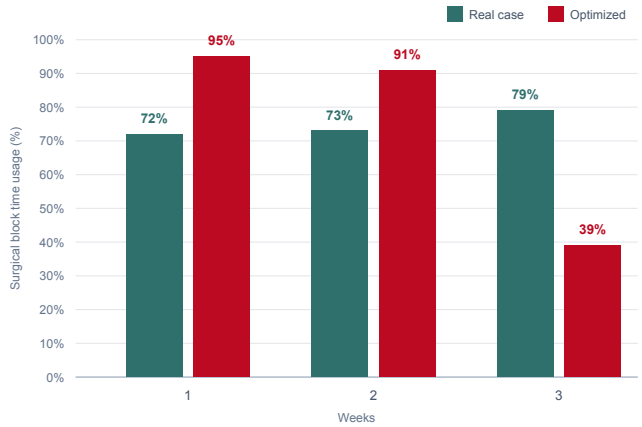

FIG. 2: Surgical Block Time Utilization: Real Case vs. Optimized Schedule.

C. Simulation Results

The benchmarks reflect the operational context of the deployed system. The scheduling process integrates with consultation: patients enter immediately after consultation, are assigned to their surgeon's schedule, and generate 3 feasible time slots within seconds. The system processes patients continuously, maintaining physician wait lists of 5–20 patients. This continuous-flow model prevents backlogs, making benchmark instances of 10–40 patients realistic operational scenarios (Table IV).

TABLE IV: Benchmark Data Results (mean values over 20 independent runs per configuration)

Physicians	Patients	Blocs	Events	Patients with Events	Avg per Patient	Run Time
1	10	2	30	10	3.0	0.17
1	10	3	30	10	3.0	0.25
2	10	2	30	10	3.0	0.34
2	10	3	30	10	3.0	0.45
1	20	2	60	20	3.0	0.98
1	20	3	60	20	3.0	1.89
2	20	2	60	20	3.0	3.40
2	20	3	60	20	3.0	3.94
1	40	2	98.34	40	2.74	600.0
1	40	3	111.15	40	2.77	600.0
2	40	2	100.53	40	2.50	600.0
2	40	3	111.56	40	2.77	600.0

The model was evaluated across three dimensions: patient count ($P \in \{10, 20, 40\}$), number of physicians ($\{1, 2\}$),

and operating room blocks ($\{2, 3\}$). For each configuration, up to three feasible time slots per patient were generated, yielding 30–111 total candidate events. To assess computational robustness, each configuration was solved 20 times with different random seeds for incoming patient characteristics while maintaining the baseline schedule structure; all reported metrics (events, average options per patient, runtime) represent means across these 20 replications (Table IV).

Key findings: (1) For operational scales ($P \leq 20$), mean solution times remain below 4 seconds across all configurations, enabling real-time consultation-integrated workflow (Figure 4). (2) For larger instances ($P = 40$) reaching the 600-second timeout, solutions remain feasible with mean optimality gaps below 15% and sufficient option generation (2.50–2.77 options per patient; Figure 3). These batch scenarios are infrequent in practice due to the continuous-flow model; 600 seconds is acceptable for offline processing. (3) The 20-run replication design demonstrates computational stability: for $P \leq 20$, optimal solutions are consistently achieved within seconds.

The framework's second phase guides patient review and slot confirmation. The MILP generates up to three feasible options per patient; using availability score σ_p, the system identifies flexible patients for early confirmation, then recomputes remaining options. This iterative approach ensures flexible patients consume non-critical options first, preserving choice for constrained patients. Patients review validated options ranked by clinical desirability, enabling shared decision-making that considers personal and occupational constraints.

The computational performance aligns with the system's consultation-driven design. Instances with $P \leq 20$ solve optimally in real-time; larger instances ($P = 40$) reach time limits but produce solutions with 15% optimality gap for infrequent batch scenarios. The limited event pool (3 options per patient) maintains tractability while preserving patient choice.

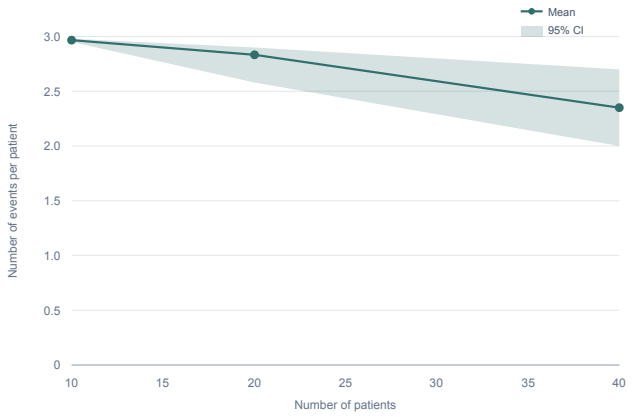

FIG. 3: Average slots per scheduled patient

V. CONCLUSION

This patient-centered operating room (OR) scheduling framework balances operational optimization with patient autonomy and shared decision-making. Implemented in the orthopedic department at CHU Montpellier (France), the system demonstrates computational feasibility and operational benefits of patient-centered surgical scheduling.

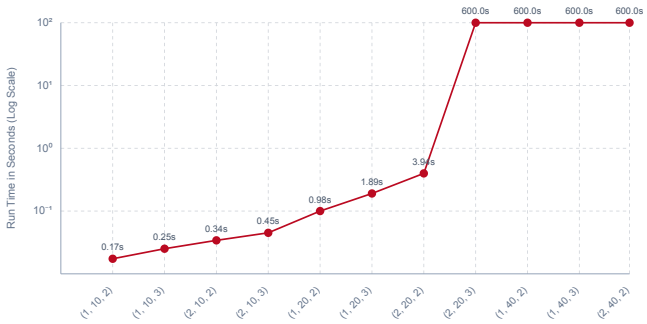

FIG. 4: Run time per instance (line plot with logarithmic scale)

The solution uses a MILP with hierarchical objectives to generate up to three time slot options for each patient. A flexibility-first heuristic prioritizes patients with flexible schedules based on early selection and availability scores while preserving options for those with constrained schedules. Patients actively participate in their scheduling decision by selecting their preferred slot option.

Benchmark simulations across 10–40 patients, 1–2 physicians, and 2–3 operating rooms demonstrate performance across diverse scenarios. Real-world validation (134 patients, 3 weeks) achieved 72–73% to 91–95% utilization increases and 36 net operations gained versus 11 in the real case. For scales $P \leq 20$, solutions arrive in under 4 seconds, enabling real-time workflow. The weighted hierarchical objective balances OR capacity with shared decision-making.

The framework was developed for elective orthopedic surgery, where scheduling flexibility is higher. Validation in other surgical specialties and at larger institutional scales is necessary.

Future work will address stochastic elements, machine learning for preferences, and real-time rescheduling. Genetic algorithms may enhance scalability for larger institutional deployments.

Operational excellence and patient care integrate through thoughtful optimization design, demonstrating that patient preferences enhance healthcare operations management.

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